MILITARY SPECIFICATION SHEET

FIXTURES, LIGHTING; FLUORESCENT, GENERAL LIGHTING

20 WATTS, 120 VOLTS, 60 HERTZ, WATERTIGHT, SYMBOLS 331.1, 331.2, 347.2 AND 347.3

This specification sheet is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the fixtures described herein shall consist of this document and the latest issue of MIL-F-16377(SH).

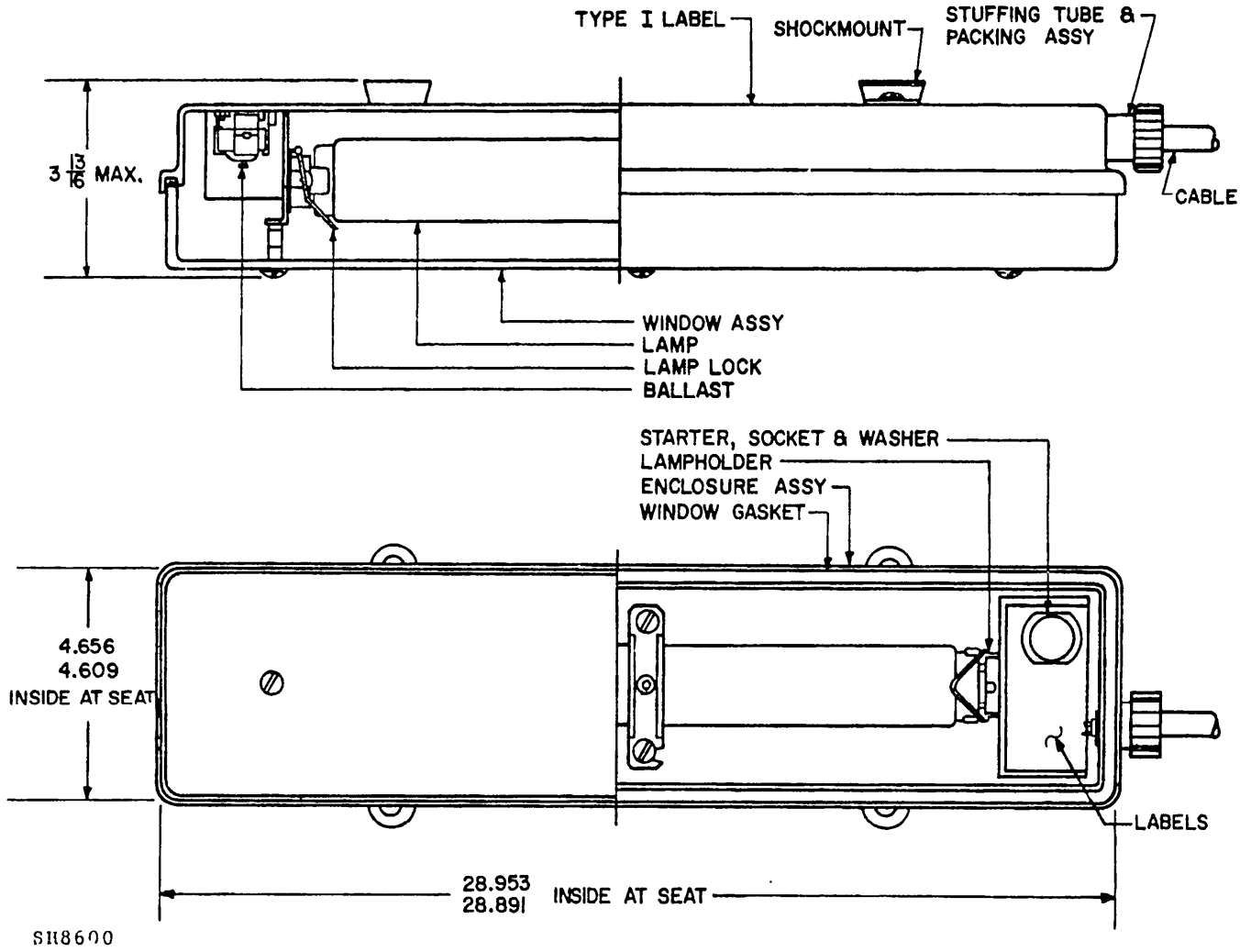


FIGURE 1. Typical view of Symbols 331.1 and 347.2.

All dimensions in inches.
For tolerances, other than specified herein, see paragraph 3.5.2 of MIL-F-16377.

FSC 6210

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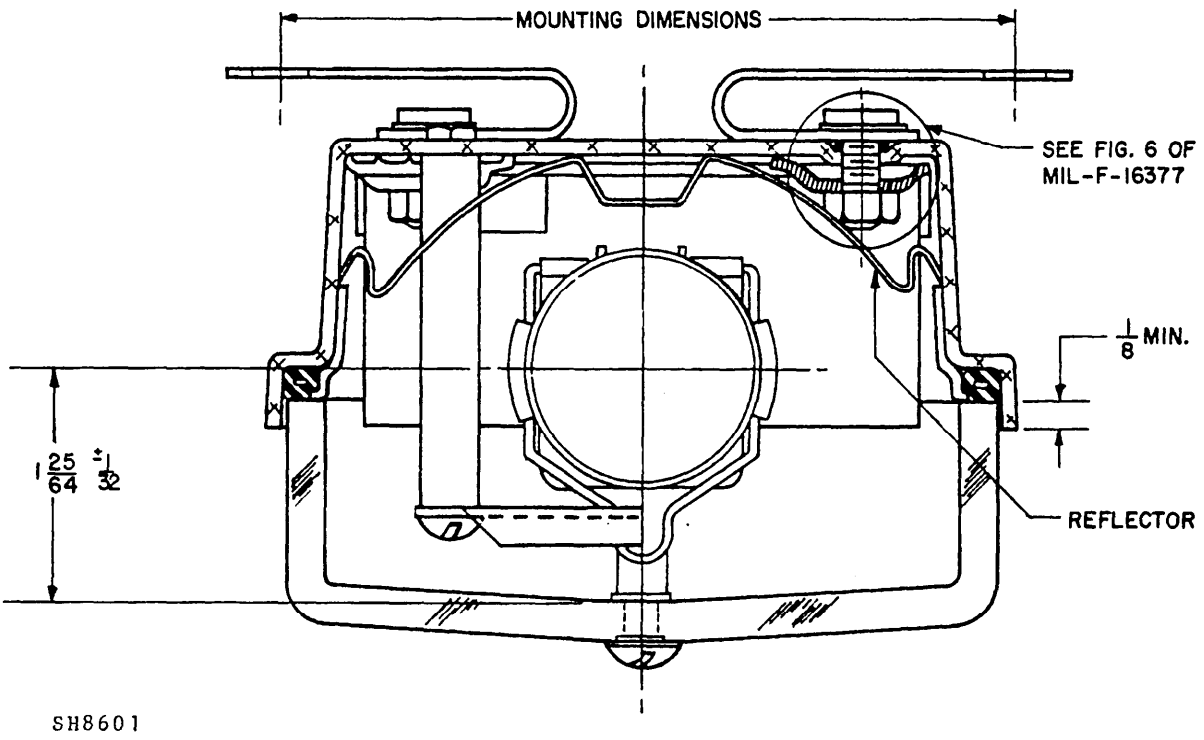


FIGURE 2. Typical section for Symbols 331.1 and 347.2.

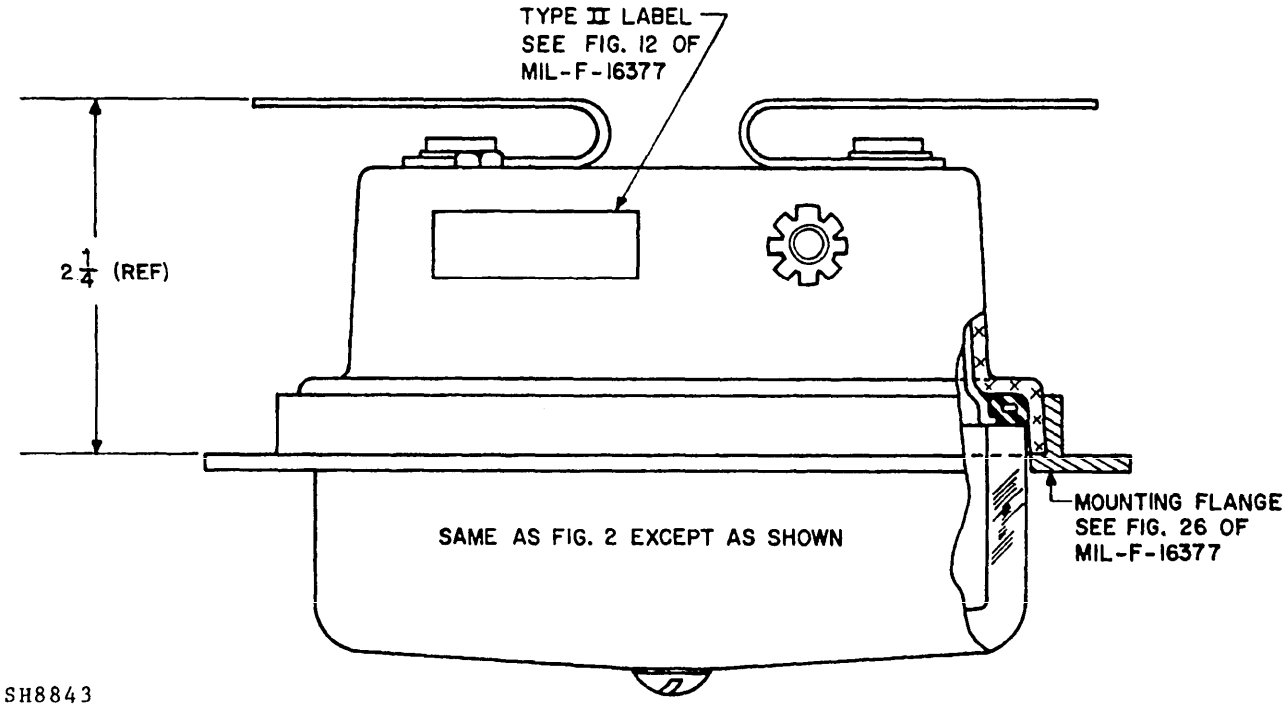
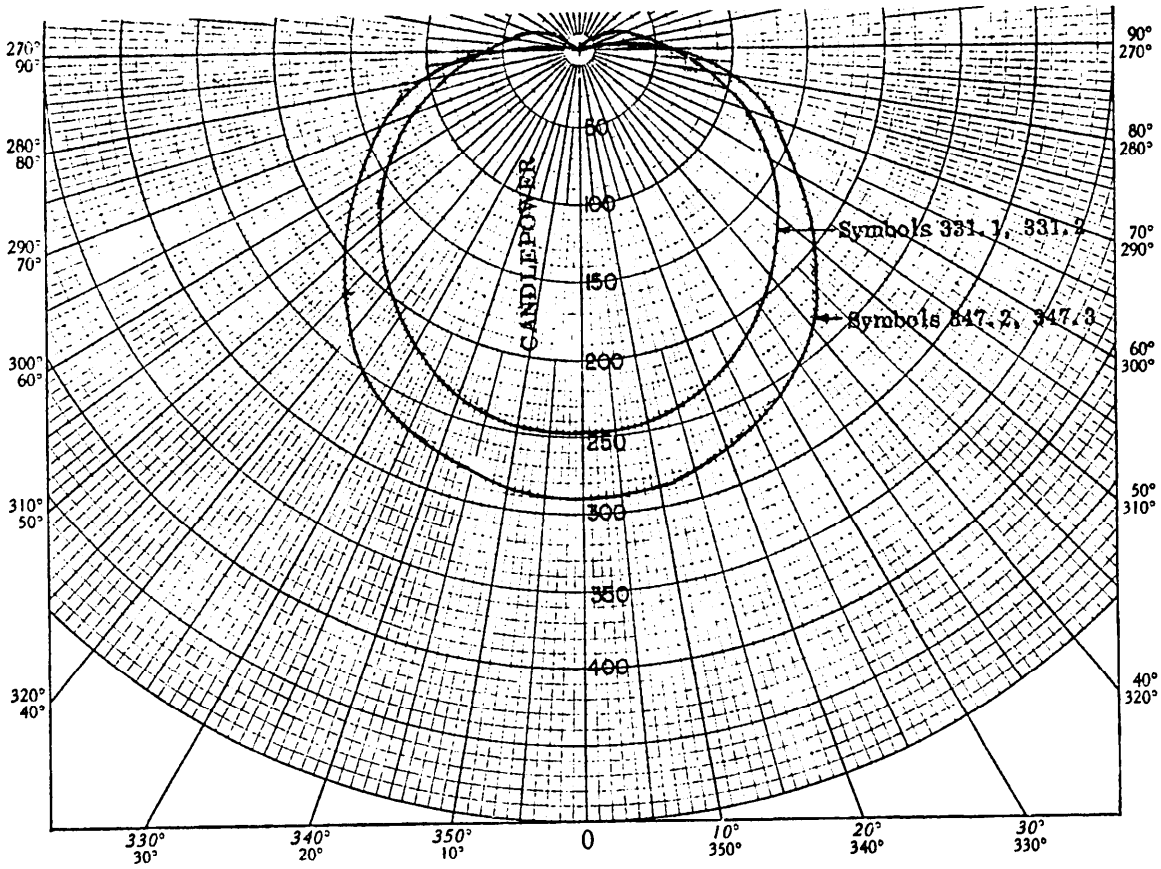


FIGURE 3. Typical section for Symbols 331.2 and 347.3.

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ANGLE	CANDLEPOWER SYMBOL NUMBER	
	331.1, 331.2	347.2, 347.3
0	243	285
5	242	285
10	240	282
15	238	278
20	230	270
25	225	260
30	216	255
35	205	250
40	194	235
45	182	218
50	170	200
55	160	185
60	145	168

ANGLE	CANDLEPOWER SYMBOL NUMBER	
	331.1, 331.2	347.2, 347.3
65	130	150
70	117	140
75	100	125
80	85	100
85	70	78
90	60	50
95	48	40
100	44	20
105	38	10
110	33	0
115	25	0
120	25	0
125	0	0

SH8604

Photometric test data:

Distance: 10 feet.
Plane: At right angles to long axis of fixture through geometric center of lamp.
Voltage: 115 volts, 60 Hz.
Ambient temperature: 76°F.
Lamps: Use lamps furnished with fixtures seasoned for at least 100 hours.
Ballast: Use ballast furnished with fixture.
Lamp correction: Distribution curves based on 1280 lumens per lamp.

FIGURE 4. Candlepower distribution curves.

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REQUIREMENTS:

CLASSIFICATION:
Type: 1
Class: 2

SYMBOL NUMBERS:
331.1: Complete lighting fixture assembly, standard mounting, with one 20-watt cool white fluorescent lamp and white diffusing window assembly.
331.2: Complete lighting fixture assembly, flush mounting with one 20-watt cool white fluorescent lamp and white diffusing window assembly.
347.2: Complete lighting fixture assembly, standard mounting with one 20-watt cool white fluorescent lamp and clear and white prismatic window assembly.
347.3: Complete lighting fixture assembly, flush mounting, with one 20-watt cool white fluorescent lamp and clear and white prismatic window assembly.

DESIGN:
Electrical rating: 20 watts, 120 volts, 60 Hertz.
Dimension and configurations: See figures 1, 2 and 3 and figure 10 of MIL-F-16377.
Photometric data: see figure 4.
Lampholder spacing: see figure 21 of MIL-F-16377.
Wiring diagram: see figure 22 of MIL-F-16377.
Mounting dimensions:
Symbols 331.1 and 347.2: see figure 25 of MIL-F-16377.
Symbols 331.2 and 347.3: see figure 26 of MIL-F-16377.
Window assembly:
Symbols 331.1 and 331.2: Military Part Number M16377/8-001, figure 27 of MIL-F-16377.
Symbols 347.2 and 347.3: Military Part Number M16377/8-003, figure 27 of MIL-F-16377.
Enclosure: Watertight. See 3.5.11 of MIL-F-16377.
Cable: 10 feet of type SJO or SJT, construction in accordance with J-C-580, except AWG size 16, two conductor (white and black).
Weight:
Symbols 331.1 and 347.2: 8 lbs. 2 oz. maximum.
Symbols 331.2 and 347.3: 9 lbs. 12 oz. maximum.
Designation and marking: The following labels shall be furnished in accordance with MIL-F-16377 applied as shown in figures 1 and 3.

(a) Two type I
(b) One type II for Symbols 331.2 and 347.3 only.
(c) One type IV. The information to be shown on the type IV label shall be "USE THE FOLLOWING REPAIR PARTS ONLY." List all standard parts shown in Table I and window assemblies of table IV by name and stock number only.

Standard Parts: The standard parts shown in table I shall be used in the construction of the fixtures specified herein.

TABLE I. Standard parts.

Name	Specification or Military part Number	National Stock Number (NSN)	Remarks
Lamp	W-L-116/16	6240-00-152-2996	Industry Number F20T12/CW
Lamp Lock	M16377/40-002	6250-00-399-7040	
Lampholder	MIL-L-970/11	6250-00-284-0481	
Startersocket	MIL-L-970/13	6250-00-295-2738	
Starter	8W , Type III of W-S-755	6250-00-299-2884	Commercial Designation: FS2
Ballast	M16377/44-001	6250-00-569-9502	
Shockmount	M16377/41-009 or M16377/41-010	6210-00-973-1747 6210-00-973-1748	See 3.5.16 of MIL-F-16377
Red filter tube	M16377/42-003	6210-00-295-2838	Not furnished See 3.7.7 of MIL-F-16377

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QUALITY ASSURANCE:

First article inspection:
Complete fixtures: See table II
Window assemblies: See table III
Quality conformance:
Comparison inspection: Same as the first article inspection
Inspection of product for delivery:
Examination: See MIL-F-16377
Examination for preparation of delivery: MIL-F-16377
Quality conformance tests: See MIL-F-16377

GENERAL INFORMATION:

Assignment of Military part numbers and general information is shown in table IV.

TABLE II. First article and comparison inspection for complete fixtures.

Tests	Requirement paragraph	Test paragraph	Remarks
Examination	3.13	4.5	
Operation	3.13	4.8.1	
Dielectric withstanding voltage	3.6.15	4.8.2	
Insulation resistance	3.6.16	4.8.3	
Enclosure effectiveness	3.5.11	4.8.14.4	Watertight
Shock	3.5.12	4.8.7	
Vibration	3.5.13	4.8.8	
Enclosure effectiveness	3.5.11	4.8.14.4	Watertight
Dielectric withstanding voltage	3.6.15	4.8.2	
Insulation resistance	3.6.16	4.8.3	
Noise	3.5.14	4.8.9	
Continuity of grounding	3.6.13.4	4.8.12	
Optical uniformity	3.7.4.1.1	4.8.4	
Stress relief	3.8.5.3	4.8.5	
Light output	3.7.1	4.8.6.1	Candlepower
Magnetic permeability	3.4.8	4.8.18	

TABLE III. First article and comparison inspection for window assemblies.

Tests	Requirement paragraph	Test paragraph	Remarks
Examination	3.13	4.5	
Optical uniformity	3.7.4.1.1	4.8.4	
Stress relief	3.8.5.3	4.8.5	
Shock	3.5.12	4.8.7	
Vibration	3.5.13	4.8.8	
Enclosure effectiveness	3.5.11	4.8.14.4	Watertight
Light output	3.7.1	4.8.6.1	Candlepower

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TABLE IV. Part number and general information.

Military Part number 1/ M16377/8-	Item name	National Stock Number (NSN)	Remarks
331.1	Complete fixture, Symbol 331.1	6210-00-548-0598	2/
331.2	Complete fixture, Symbol 331.2		2/
347.2	Complete fixture, Symbol 347.2		New
247.3	Complete fixture, Symbol 347.3		New
001	Window, white diffusing	6210-00-548-0601	3/
003	Window, clear and white prismatic		New

1/ Military part numbers M16377/8-347, 347.1 and 002 have been superseded by Military part numbers M16377/8-347.2, 347.3 and M16377/67-001 respectively.

2/ Formerly covered on NAVSHIPS Drawing 9000-S6401-74482 and MS17194.

3/ Formerly covered on MS16891-40.

Revision letters are not used to denote changes due to the extensiveness of the changes.

Preparing activity:
Navy - SH
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